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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Adaptive Blood Donor Matching System	
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Abstract

AI-Adaptive Blood Donor Matching System is a real-world application designed to help hospitals and blood banks quickly identify suitable blood donors during emergency situations. The system uses **Artificial Intelligence (AI)** and intelligent matching techniques to analyze patient blood requirements and donor information such as blood group, location, availability, and donation history.

The system provides a **donor compatibility and priority score** and recommends the most suitable available donors based on the urgency of the blood request. It can also adapt to changing requirements such as donor availability, emergency priority, and location to improve the efficiency of the matching process. The system can notify eligible donors and help reduce the time required to find suitable blood.

The application can be used by **hospitals, blood banks, healthcare organizations, and registered donors** to improve emergency blood coordination. By combining **AI, adaptive decision-making, real-time data, and automated donor recommendations**, the project aims to reduce delays in emergency blood requests and improve the overall efficiency of blood donation management.

Signature of the Guide HOD